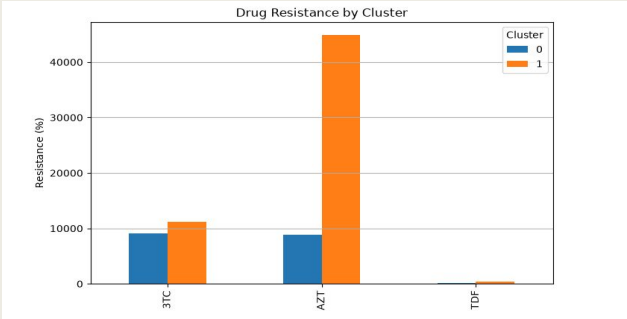
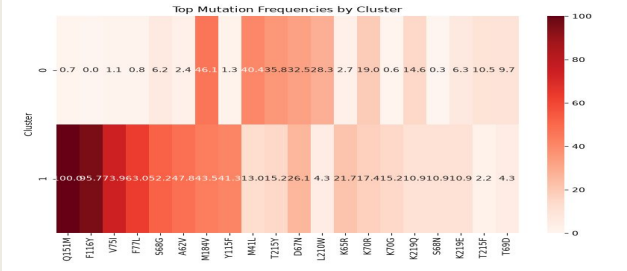
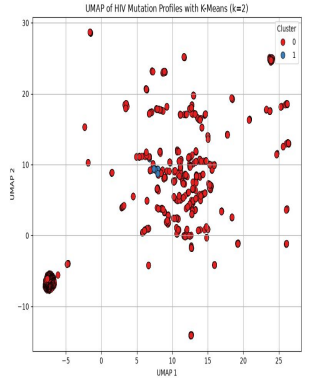
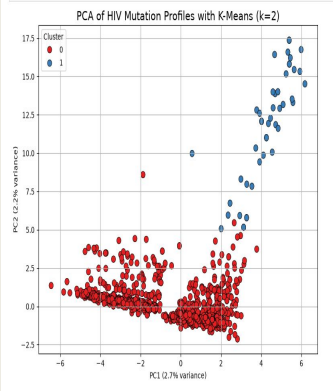


Explanatory analysis on Stanford datasets using PCA-K-means and UMAP analysis.



Overview

Dataset

Mutation features extracted from **Stanford datasets for HIV** reverse transcriptase isolates

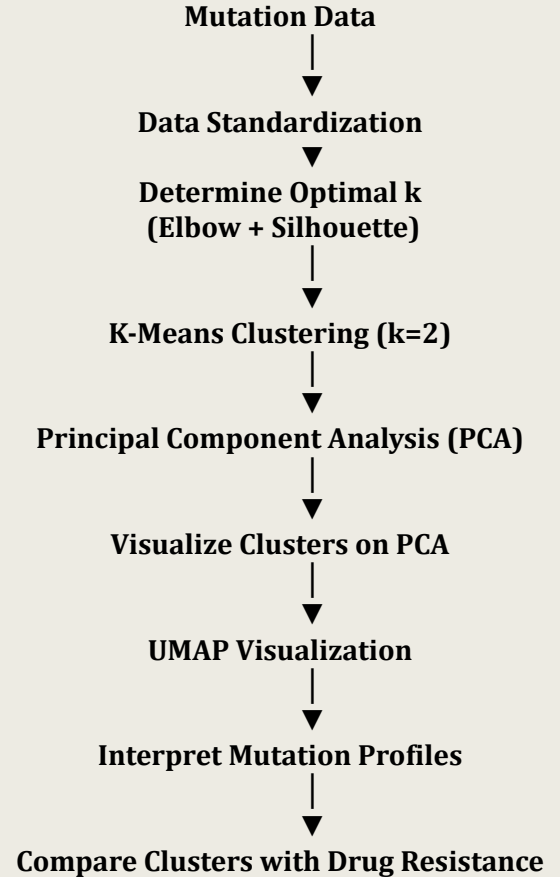
Approach

Unsupervised machine learning to identify natural clustering patterns without prior labels

Goal

Foundation for supervised drug-resistance prediction models using cluster insights

Complete Analysis Workflow



Optimal Cluster Identification

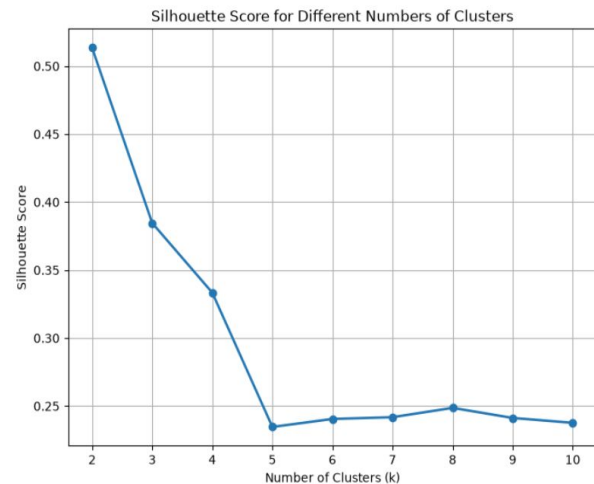
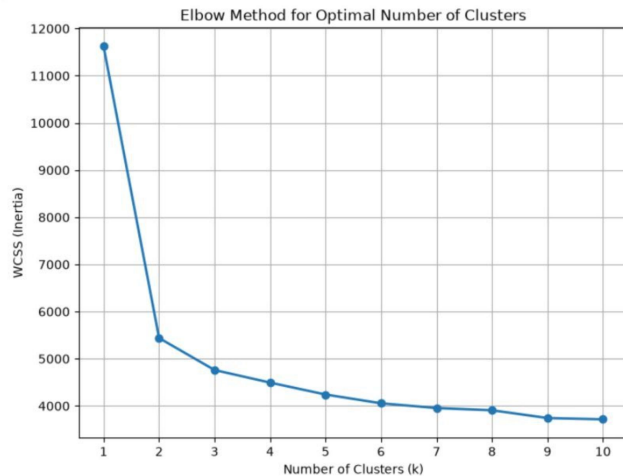
Elbow Method

The within-cluster sum of squares (inertia) curve shows a clear inflection point at **k = 2**, indicating that adding further clusters yields diminishing returns in explanatory power.

Silhouette Score Validation

A silhouette score of **0.51** at $k = 2$ confirms moderate-to-strong cohesion within clusters and separation between them — the highest score across all tested values of k .

Both independent metrics converge on $k = 2$ as the statistically optimal solution.



Cluster Validation

K-Means Partitioning

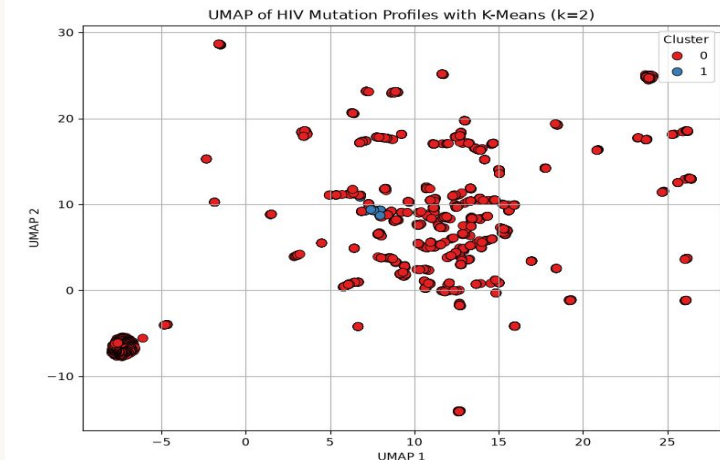
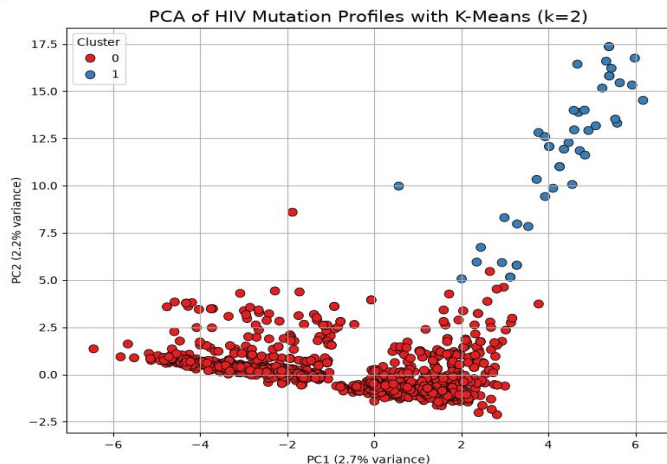
K-Means clustering successfully partitioned HIV isolates into two distinct, non-overlapping groups based on mutation feature profiles.

PCA Visualization

Principal Component Analysis reveals clear spatial separation between clusters along PC1 and PC2, confirming structural distinctness in reduced dimensions.

UMAP Confirmation

UMAP dimensionality reduction independently confirms the two-cluster topology, reinforcing that the separation reflects genuine biological structure.



Top Mutations in Each Clusters

Rank	Top Mutations in Cluster 0	Frequency (%)	Top Mutations in Cluster 1	Frequency (%)
1	M184V	46.07	Q151M	100.00
2	M41L	40.38	F116Y	95.65
3	T215Y	35.78	V75I	73.91
4	D67N	32.49	F77L	63.04
5	L210W	28.26	S68G	52.17
6	K70R	18.95	A62V	47.82
7	K219Q	14.59	M184V	43.47
8	T215F	10.49	Y115F	41.30
9	T69D	9.71	D67N	26.08
10	E44D	8.58	K65R	21.73
11	L74V	7.76	K70R	17.39
12	K219E	6.26	K70G	15.21
13	S68G	6.18	T215Y	15.21

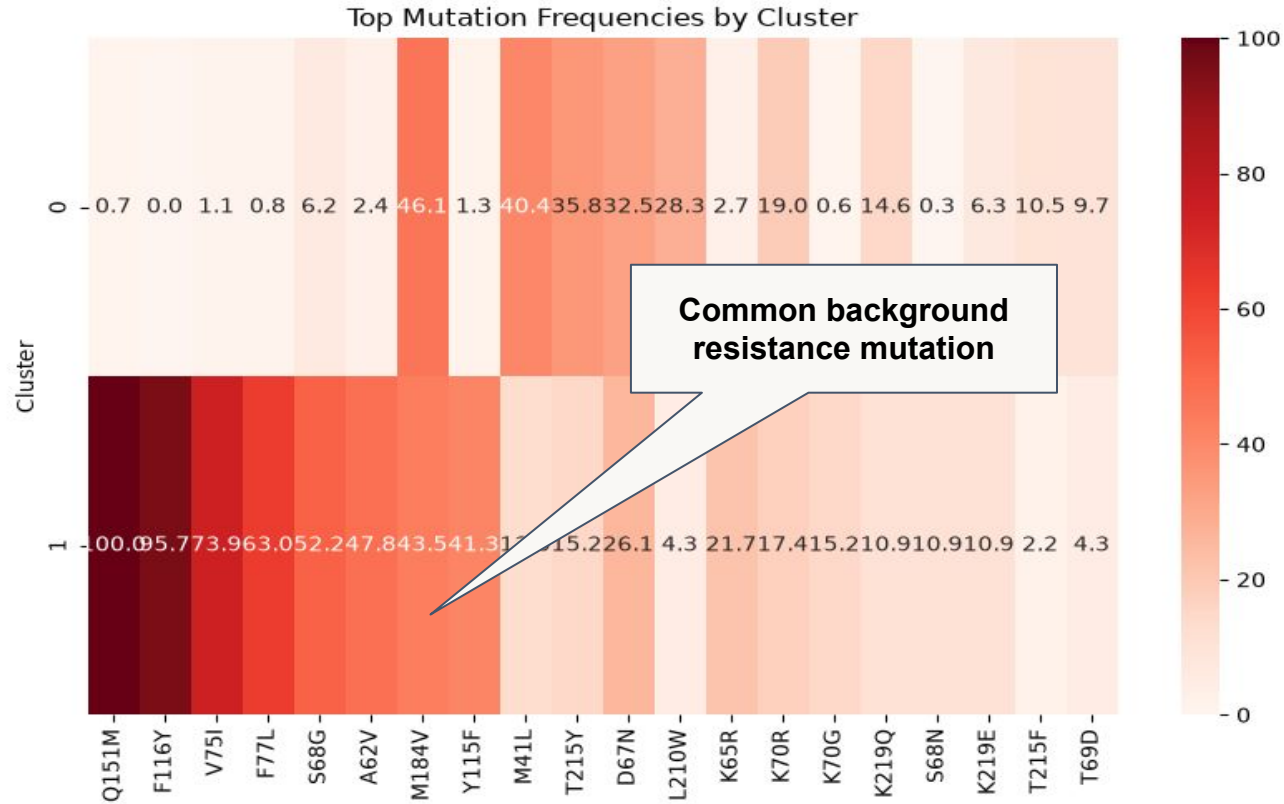
Overall Pathway Distribution

Drug Resistance pathway	Cluster 0	Cluster 1
TAM*-1	★★★★★★ Dominant	★ Low
TAM-2	★★★★★ Dominant	★★ Moderate
M184V pathway	★★★★ High	★★★★ High
K65R pathway	★ Very low	★★★★ High
Q151M complex	Not detected	★★★★★★ Dominant
Codon 69 insertion	Not detected	Not detected

* Thymidine Analog Mutations



Top Mutation Frequencies by Cluster Heatmap



Drug Resistance Association

Chi-square statistical testing was applied to evaluate whether cluster membership predicts drug resistance phenotype. All four tested drugs yielded **highly significant associations**.

3TC (Lamivudine) — $p < 0.001$

Cluster membership is significantly associated with lamivudine resistance, driven primarily by the M184V mutation prevalent in Cluster 0.

AZT (Zidovudine) — $p < 0.001$

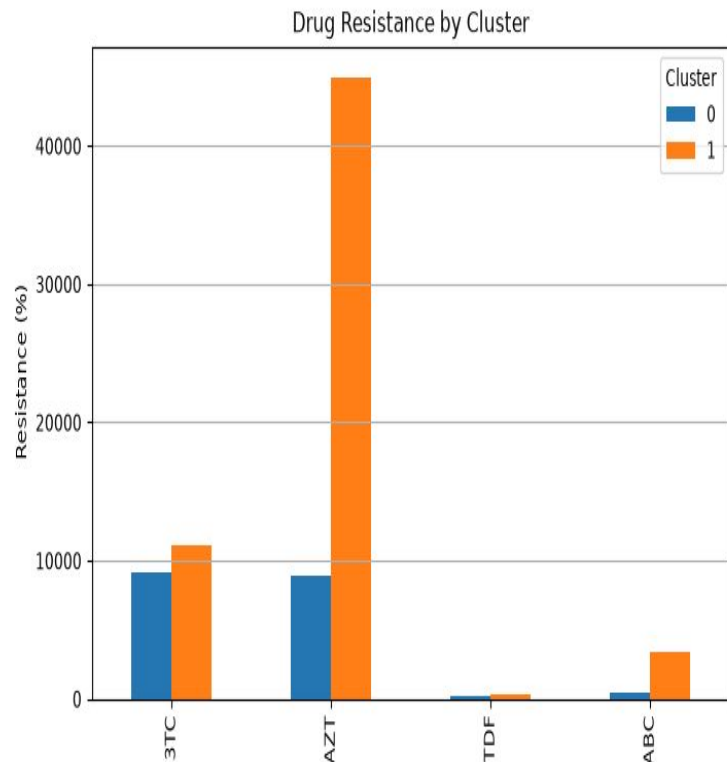
Thymidine analog mutations (M41L, T215Y, D67N) in Cluster 0 produce strong AZT resistance linkage to cluster assignment.

TDF (Tenofovir) — $p < 0.001$

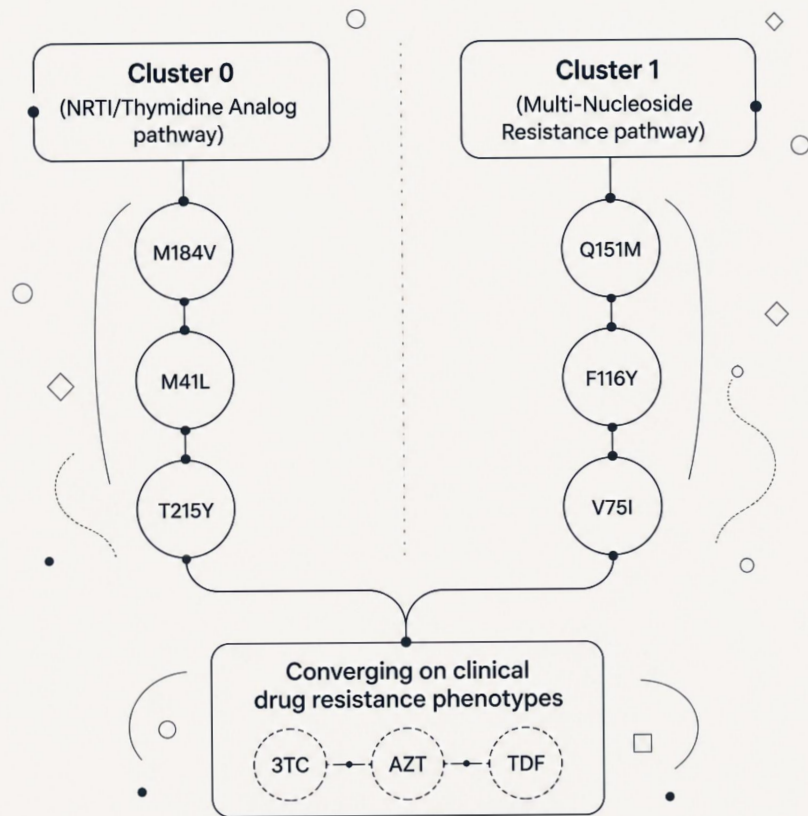
Tenofovir resistance is also significantly differentiated by cluster

ABC (Abacavir) — $p < 0.001$

Cluster 1 contains a higher frequency of **K65R**, which is consistent with its significant association with ABC resistance.



Biological Significance



Two Distinct Clusters

The two clusters represent genuinely separate biological routes to NRTI resistance.

Phenotype-Genotype Concordance

Similar mutation profiles within each cluster correlate with shared drug-resistance phenotypes, confirming that genotypic grouping captures clinically meaningful distinctions.

Validation

The fact that an unsupervised algorithm — with no resistance labels provided — recovered clinically interpretable based on this **Explanatory Analysis**.